

SPARQL

Query for RDF (Ontology and Knowledge Graph)

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Recaps

- Ontology and Knowledge graph structure

- Form in Triple

[subject]	predicate	[object]
[ex:Alan]	ex:studiesAt	[ex:UniversityA]
[ex:Bob]	ex:studiesAt	[ex:UniversityA]
[ex:UniversityA]	ex:locatedIn	[ex:Bangkok]
[ex:Alan]	ex:studies	[ex:ComScience]
[ex:Bob]	ex:studies	[ex:Linguistics]
[ex:ComScience]	ex:part of	[ex:FacultyOfScience]
[ex:Linguistics]	ex:partOf	[ex:FacultyOfHumanities]

- Prefix and URI - a marker to define the entities

- ex: → example and URI → Alan
- Combine to ex:Alan

What is SPARQL?

- SPARQL (SPARQL Protocol and RDF Query Language) is a query language for RDF-based knowledge graphs.
- It allows users to retrieve and manipulate information by specifying patterns of relationships between resources, rather than querying rows and columns in tables.
 - selects specific data from a data store
 - uses variables to represent unknown values
 - supports filtering and sorting
 - supports joins/combining information
 - supports aggregation such as COUNT, SUM, and AVG

What is WikiData?

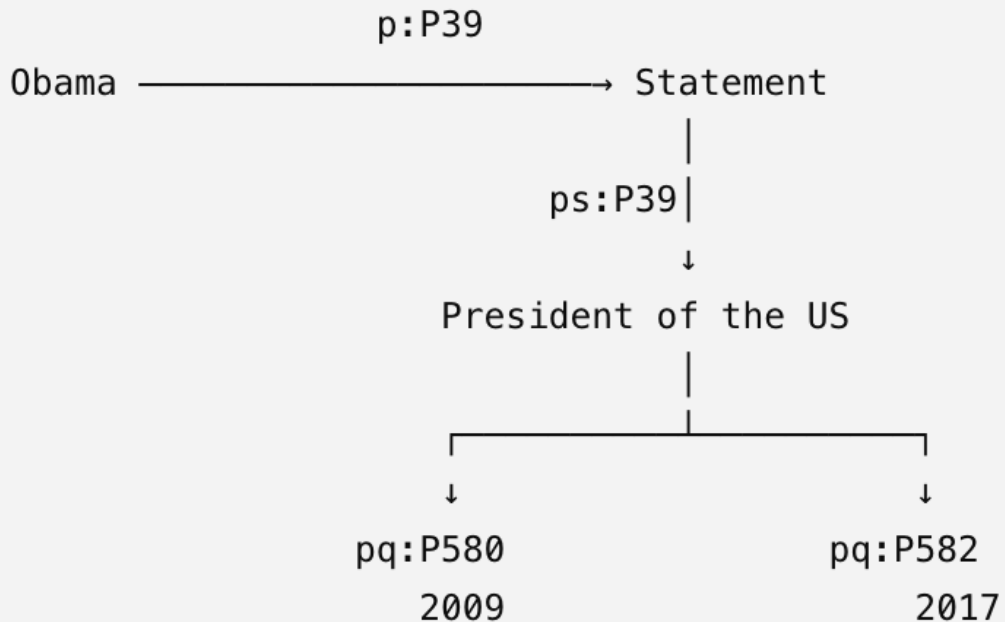
- **Wikidata** is a free, collaborative knowledge graph operated by the Wikimedia Foundation. It stores structured information about entities such as people, places, organisations, works, scientific concepts, and events.
- Unlike Wikipedia, which primarily presents information written for humans, Wikidata stores information in a machine-readable form using structured statements.

WikiData Details

Type	List
Prefix	For Entity → wd: http://www.wikidata.org/entity/ For Predicate → wdt: http://www.wikidata.org/prop/direct/ For Statement • p: → statement • ps: → statement's main value • pq: → qualifier
URI	Q for entity following with numeric ID • Q869 → Thailand • Q1861 → Bangkok P for predicate following with numeric ID • P36 → capital
Label	en → English (such as Thailand, Bangkok) th → Thai (such as ไทย, กรุงเทพฯ)

Prefix and Statements

wd:Q76	wdt:P39	wd:Q11696
Obama	position held	President of the United States
	*Statement	
	ps:P39 → President of the United States	
	[start time] pq:P580 → 2009	
	[end time] pq:P582 → 2017	



wd: + wdt: "Obama is President"

p: + ps: + pq: "Obama was President from 2009 to 2017"

➤ Additional Commands in Wikidata Query

- For retrieving label
 - `SERVICE wikibase:label { bd:serviceParam wikibase:language "AUTO_LANGUAGE], en" . }`
 - `[AUTO_LANGUAGE]`
 - English → en, Thai → th
 - Variable `?x` → `?xLabel`
- For limit queried item
 - `LIMIT n` → `LIMIT 20` , `LIMIT 100`
- For sorting
 - `ORDER BY DESC(?x)`
 - `ORDER BY ASC(?x)`

Do it yourself

1. Find airports in Asia and display their geographic coordinates. Visualize the results on a map.
2. Find cities in Europe with a population greater than 1 million. Sort from highest to lowest population and return a maximum of 100 cities.
3. Find museums around the world and display their genre and established date as optional.
4. Find scientists who received a Nobel Prize after 2000. Display the scientist's name, nationality, Nobel Prize received, and the year the prize was received. Sort the results by the award year from earliest to latest.
5. Find mountains in Thailand with an elevation greater than 1,000 meters. For each mountain, display its name and elevation. Also display the province or administrative area where the mountain is located, if this information is available (optional). Sort the results by elevation from highest to lowest.
6. Find people who have held the position of Prime Minister of Thailand. Display their name, start date, end date if available, how many days they were in office, age at the start of their term, and age at the end of their term. Sort by start date from earliest to latest.